Xi'an Jiaotong-Liverpool University
西交利物浦大学

CPT208 Human-Centric Computing

Coursework: Project

Module Name:	CPT208 Human-Centric Computing	
Module Leader:	Dr Yue Li and Dr Teng Ma	
Deliverables and Due Dates:	Poster Presentation Portfolio, System & Video Demo	**Week 8** **Week 10**
Marks:	40% of the module mark	

Learning Outcomes Addressed

A. Apply a user-centered approach to interaction design, emphasizing the importance of supporting equality, diversity, and inclusion in technical work.
B. Create, develop, and deploy interactive computing systems, incorporating legal, social, ethical, and professional considerations to achieve specific business objectives.
C. Critically assess and evaluate the usability and user experience of interactive systems using established criteria.

Submission Format

- **Only one submission** is needed from each group.
- Naming formats and examples:
  - CPT208_GroupA1-1_ProjectName_Poster.pdf
  - CPT208_GroupA1-1_ProjectName_Portfolio.zip
  - CPT208_GroupA1-1_ProjectName_DemoSystem.zip
  - CPT208_GroupA1-1_ProjectName_DemoVideo.mp4

Artificial Intelligence Permissions

For the **Poster** submission, students may make **non-substantive use of AI**. Students are permitted to use Generative AI tools specifically for visual asset generation and grammatical refinement.

<u>Permitted AI Uses for Posters</u>

- Image Generation: You may use AI to create illustrative backgrounds, icons, or conceptual visuals that help communicate your "Playful Experience".
- Grammar Editing: You may use AI for minor tasks such as grammatical corrections or improving the clarity of your text.

The use of AI tools beyond the scope outlined above would constitute a breach of the University Academic Integrity Policy and will result in penalties, which may include an overall mark of zero for the assignment and a formal notification of academic misconduct. If you are unsure on whether a specific use of AI is allowed or not, please consult the module leader.

Strict Requirements

- Non-Substantive Role: AI must not be used to define your project motivation, research findings, or core design logic. These must remain your group's original work.
- Acknowledgment: Every AI-generated image must be accompanied by a small caption or acknowledgement (e.g., "Image generated via [Tool Name]").
- Mandatory Citation: You must include a formal citation in the References section of your Process Portfolio.

Citation Example

In your reference list, please use the following format:

[1] Gemini Nano Banana, version 1.0, accessed on 2026-02-24, available at https://gemini.google.com. Used for generating the conceptual hero image in the illustrative figure.

For the **Portfolio, System & Video Demo** submissions, students may make **substantive use of AI**, including advanced AI integration to help you build functional web-based prototypes. However, the core design logic and the "human-centric" justification must be your own.

Permitted AI Uses for System & Video

- Vibe Coding & Assisted Scripting: You may use LLMs to generate code snippets, debug logic, or scaffold your web application.
- Video Post-Production: You may use AI for generating voiceovers, cleaning audio, or assisting with video transitions to achieve a professional finish.

The use of AI tools beyond the scope outlined above would constitute a breach of the University Academic Integrity Policy and will result in penalties, which may include an overall mark of zero for the assignment and a formal notification of academic misconduct. If you are unsure on whether a specific use of AI is allowed or not, please consult the module leader.

Mandatory Requirements

- Reflection in Portfolio: If you use "vibe coding" to build your system, you must include a "Technical Reflection" section in your Process Portfolio. You must explain:
    - What prompts were used to generate the core logic.
    - How you verified the code met your user requirements.
    - Any ethical considerations regarding AI-generated code (e.g., accessibility or bias).
- Mandatory Citation: Every tool used must be documented in your portfolio's reference list.

Citation Example

In your portfolio's reference list, please use the following format:

[1] Claude 3.5 Sonnet, v1.0, accessed on 2026-02-24, available at https://claude.ai. Used for refining the user persona descriptions and generating the initial storyboard script for the Video Demo.
[2] Cursor, v0.40, accessed on 2026-02-24, available at https://cursor.com. Used for vibe coding the system's backend API and debugging the CSS layout for mobile responsiveness.

Task Descriptions: **Playful** Experience Design

Summary of Key Information

Design Themes

A. **Digital Heritage & Play:** Creating playful onsite experiences for Suzhou landmarks (e.g., Chang Gate or Maple Bridge).
B. **Social Connectivity:** Tools to facilitate meaningful campus discussions or bridge the generation gap.
C. **Active Lifestyles:** Gamified systems for runners, climbers, or gym-goers.

Deliverables

1. **Poster Presentation (40%, Week 8):** Presentation
2. **Process Portfolio, System & Video (60%, Week 10)**
   - **Portfolio:** A hosted web page details the process (github.io).
   - **System:** A functional web-based prototype, ideally via a live URL.
   - **Video:** A 2-minute "CHI-style" teaser showcasing the user journey and interaction.

Technical Constraints

- **Platform:** Must be a Web App (ideally accessible via URL).
- **Tools:** Use free-tier services only (e.g., Figma for UI, VS Code for dev, GitHub for hosting).
- **AI Policy:** AI may be used for coding assistance or asset generation, but **all** design decisions and user research must be primary data collected by the group.

Suggested Free Tools

- **Portfolio:** GitHub Pages.
- **Design:** Figma.
- **Development:** VS Code, GitHub Desktop.
- **Web App Hosting:** GitHub Pages or Vercel.
- **Video:** CapCut.

You will prepare and present your group project **as a group of four**. The grouping will open during the seminar in **Week 1**. If you have not registered for a group until the seminar of **Week 2**, you will be allocated to a group.

Once you have registered for a group and topic, the change of groups or topics is **NOT ALLOWED**. Please be careful with your selection and take responsibility for your group work.

Design Themes & Topics

	Topic	Stakeholders	Device	Good to Have	Where
A1	Suzhou Grand Cannel – Chang Gate 阊门	Visitors & Residents	Mobile	XR, GenAI, Location-based	Onsite
A2	Suzhou Grand Cannel – Maple Bridge 枫桥	Visitors & Residents	Mobile	XR, GenAI, Location-based	Onsite
A3	My Stories in Museums	Visitors (Content creator & Audience)	Mobile	XR, GenAI, Location-based	Hybrid
B1	XJTLU Group Discussions	XJTLU students & Language and academic teachers	Mobile, PC	GenAI	Onsite
B2	XJTLU Student Development	XJTLU students & Development advisors	Mobile, PC	GenAI	Hybrid
B3	XJTLU Campus Tour	New students; External visitors	Mobile	XR, Location-based, Tangible	Onsite / Online
B4	Relation between Generations	Parents & Children; Grandparents & Children	Mobile	XR, Tangible	Hybrid
C1	Go Runners	Runners & Activity organizers	Mobile, PC	Wearable, Location-based	Onsite
C2	Go Climbers	Climbers & Coaches	Mobile	XR, Wearable, Location-based	Onsite
C3	Go Trainers	Gym trainers & Instructors; Other sports players	Mobile	XR, Wearable, Location-based	Onsite
D	*Open topic*				

You should research extensively (from both academic literatures and commercial products) on the given topic and consider the following:

1. Who are the stakeholders / users of the system?
2. What does the system allow users to do?
3. Are there any issues with the existing system(s)?
4. Are there any user requirements that are not supported with the existing system(s)?
5. What are the design alternatives to solve the existing issues / fulfill user requirements?
6. How can you make prototypes of the design alternatives?
7. How can you evaluate your design?
8. What are the implications of your design?

Deliverables

Throughout the semester, your will incrementally build up a process portfolio that records your design process. Below are some key milestones for submissions:

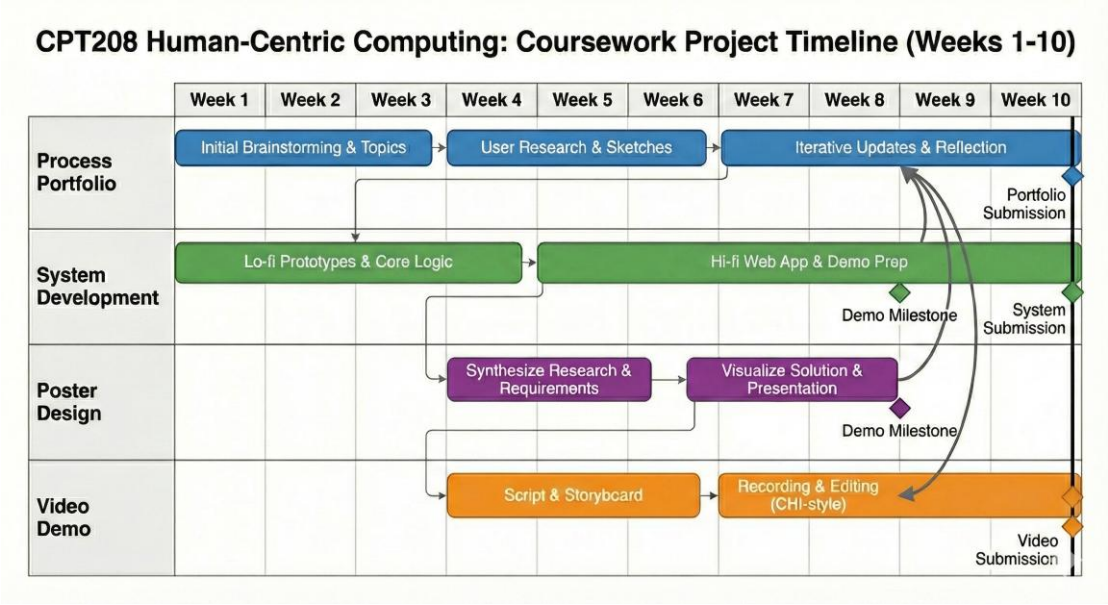

Figure 1. Suggested project timeline showing the key milestone submissions.

(Image generated by Gemini with Nano Banana Pro)

To clarify & approach the tasks,

- **Portfolio** is a detailed documentation of the design process, updated throughout the semester.
  - W1-W3**:** Record initial brainstorming, group roles, and topic selection.
  - W4-W6: Document user research findings, personas, and early sketches.
  - W7-W10: Log iterative updates, evaluation results, and final reflections.
- **System** is a functional prototype demonstrating the group's proposed solution.
  - W1-W4: Low-fidelity prototyping and core feature logic.
  - W5-W8: High-fidelity web app development leading to the Week 8 Demo.
  - W9-W10: Bug fixing and usability refinements for the final submission.
- **Poster** is a mid-term milestone showcasing research, design progress, and initial prototypes.
  - W4-W5: Synthesize research gaps and design requirements.
  - W6-W8: Visualize the final solution for the mid-term presentation.
- **Video** is a concise and engaging video showcasing the system's features and impact.
  - W4-W6: Scripting and storyboarding the user journey.
  - W7-W10: Recording the functional system and editing the 2-minute CHI-style teaser.

Project Timeline & Checklist

	Deliverable	Checklist
Week 2	**Group & Topic Lock**	* 4 members registered. * Topic selected and registered on learning mall. * Process Portfolio site (GitHub) initialized.
Week 4	**User Research Conducted**	* 2 personas created. * User journey map completed. * Findings from 5+ academic papers cited in Portfolio.
Week 6	**Mid-term Poster Draft**	* Aims and objectives clearly stated. * Visuals of initial prototypes included. * Novelty and significance explained.
Week 8	**Final Poster & Alpha System**	* Functional core logic (web-based) realized in System. * Initial user testing data recorded in Portfolio. * Design Alternatives compared and documented.
Week 10	**Final Submission**	* Live URL for the Web App. * 2-min Video (MP4). * Final Portfolio (including iterative feedback logs).

Instructions for the Process Portfolio

In this submission, you should present your work with a focus on the followings:

1. Demonstrate an iterative design, prototyping, and evaluation process
2. Explain the prototypes used in the design process, with comparisons of alternatives
3. Use appropriate methods to evaluate the design alternatives
4. Show a valid collection and analysis of data from the correct target user demographics with reasonable sample sizes
5. Demonstrate a good discussion on the evaluation results and design implications

Specifically, the Process Portfolio should be a window to showcase your work. You should cover the following content in your portfolio:

1. Motivation & Research

- **The Why:** A 200-word statement on why your group chose your specific track (Heritage, Social, or Active).
- **The Gap:** Summarize 4 academic papers and 4 commercial products. Instead of a summary, list **3 things they did well** and **3 things they missed**.
- **The Stakeholders:** Define primary and secondary users with distinct **Personas**.

2. User Requirements

- **User Journey Map:** A visual map showing the current pain points of your target users.
- **Requirements List:** A bulleted list of **3 must-haves** for the system to be considered "Playful".
- **Evidence of Life:** Upload 5 photos or short clip videos of your group interviewing a potential user or observing the site (e.g., Suzhou Grand Canal).

3. Ideation & Alternatives

- **The "Crazy Eights":** Photos of 8 rapid hand-drawn sketches of different UI ideas.
- **Design Alternatives:** Compare at least 2-3 different ways the system could work and explain why you chose one over the other(s).
- **Low-Fi Prototype:** A link to a clickable Figma prototype.

4. Technical Implementation

- **System Architecture:** A diagram showing how your Web App handles data.
- **High-Fi Prototype:** Ideally a URL to the hosted system (e.g., yourproject.github.io or yourproject.vercel.app).
- **Individual Contributions:** A table showing exactly what each of the 4 members built (Code, UI Design, Content, or Testing, etc.).

5. Evaluation & Reflection

- **Usability Testing:** Results from testing your "Alpha" version with 3 real people.
- **Iterative Refinement:** "Before and After" screenshots showing a change you made based on user feedback.
- **Final Reflection:** A brief discussion on the social/ethical implications of your design, and the use of AI in this project (check more details in the Generative AI Permissions section).

Instructions for the System Development

The system is the functional heart of your project. It must be a live, interactive web application that demonstrates your group's solution to the identified human-centric challenge.

Key Objectives:

1. **Functionality:** Deliver a working prototype that users can interact with via a URL.
2. **User Experience:** Demonstrate a high-fidelity interface that aligns with your design requirements.
3. **Technical Deployment:** Show proficiency in deploying to modern cloud platforms (e.g., GitHub pages, Vercel).
4. **Accessibility:** Ensure the system is usable and follows basic web standards.

Required Content:

1. **Live URL:** A direct link to your hosted web app. Ensure the site is "Public" and remains active throughout the marking period.
2. **Core Feature Set:** The system must implement at least **3 must-have** playful features identified in your Requirements List.
3. **Responsive Design:** The interface should be optimized for the device specified in your track (e.g., Mobile / PC).
4. **Source Code Repository:** A link to your group's GitHub repository.
    - **README.md:** Must include setup instructions and a list of technologies used (e.g., Three.js, React).
    - **Vibe Coding Logs:** If AI was used for coding, include a folder named /ai-logs containing the primary prompts used to generate core components.
5. **Data Handling:** Evidence of how the system manages user input or interaction states (as outlined in your Portfolio's System Architecture diagram).

Instructions for the Poster Submission

In the **poster presentation**, you should complete the following tasks:

1. Clearly state the project motivation, as well as aims and objectives
2. Clearly specify and study the target group of users
3. Research extensively and summarize the key issues on the existing works
4. Show some efforts in collecting design requirements
5. Propose effective and novel design solutions, use prototypes where possible

Poster Size: A1, 59.5 x 84.1 cm (23.4 x 33.1 inches).

Specifically, you should cover the following content in your poster:

1. Header & Brand

- **Project Title:** A catchy, playful name for your system (e.g., *"Suzhou Whispers"* instead of *"Project A1"*).
- **Team Info:** Group ID, member names and student IDs.

2. The Problem Space

- Clearly define the motivation behind your project.
- Briefly highlight what is missing in existing commercial or academic solutions.
- Identify your primary and secondary user groups using a concise Persona snapshot.

3. The Design Solutions

- State your specific aims and objectives.
- Describe your novel design concept.
- Explain what makes this experience engaging or human-centric.
- A simplified visual (e.g., user journey map) showing how a user interacts with your system to solve their pain points.

4. Prototypes & Iteration

- Show the evolution from early sketches (low-fi) to your current web-based prototype (high-fi).
- A small diagram showing how the system works (e.g., WebXR on Vercel).
- Explain one major design decision you made between 2-3 options.

5. Early Evaluation & Impact

- Summary of any pilot testing or requirements collected.
- Why does this project matter in the context of the topic or human-centric design?

6. References & Acknowledgments

- **AI Disclosure:** Include the required citation for any AI-generated images used in the poster (e.g., Gemini Nano Banana) as per university policy.
- **Academic References:** List at least 3 key papers that influenced your research.

Instructions for Video Demo Submission

The video is your "pitch" to the world. It should be a concise, engaging 2-minute demonstration that captures the vibe and impact of your system.

Format: MP4 encoding (with H.264 compression), 1080p.

For more details, check out the two guides: https://sigchi.org/resources/guides-for-authors/videos/ & https://chi2025.acm.org/guide-to-a-successful-video-submission/

Key Objectives:

1. **Narrative:** Tell a story that starts with a user's pain point and ends with your playful solution.
2. **Clarity:** Use screen recordings to show exactly how the system works.
3. **Engagement:** Use high-quality audio and visuals to maintain a standard teaser video.

Required Content:

1. **Cover:** Show a representative image of your project, with your group ID, project title, your names, and student IDs.
2. **The Hook (~10s)** Visually demonstrate the problem space (e.g., footage of a crowded site in Suzhou or a lonely student study session).
3. **The Concept (~15s):** Introduce your system and its core human-centric value proposition.
4. **The Walkthrough (~55s):** High-quality screen recordings of the High-Fi prototype in action.
    - Show a user navigating through the 3 core features.
    - Use call-outs (text overlays) to highlight specific playful interactions.
5. **The Impact (~35s):** Show clips of evaluation results or quote from your usability testing.
6. **Closing & Credits (~5s):** Acknowledge the help your received throughout the project; display the mandatory AI Disclosure if AI-generated voices or visuals were used.

Marking Criteria

Component	Marks	Marking Criteria
Poster	40	**Excellent** (36–40): Comprehensive and well-researched content, clear visuals, and innovative design solutions. **Good** (28–35): Clear and focused content, with minor gaps in research or design details. **Satisfactory** (20–27): Basic content with limited research or unclear design solutions. **Needs Improvement** (0–19): Lacks clarity, research, or meaningful design progress.
Portfolio	20	**Excellent** (18–20): Clear, well-organized, and detailed specification with evidence of iteration and refinement. **Good** (15–17): Well-structured and mostly complete, with minor gaps in detail or clarity. **Satisfactory** (10–14): Basic content with limited updates or unclear descriptions. **Needs Improvement** (0–9): Incomplete or poorly organized specification.
System	20	**Excellent** (18–20): Fully functional system with innovative features, robust implementation, and excellent usability. **Good** (15–17): Mostly functional system with minor issues or missing features. **Satisfactory** (10–14): Partially functional system with limited features or usability issues. **Needs Improvement** (0–9): Non-functional system or significant gaps in implementation.
Video	20	**Excellent** (18–20): Highly engaging, professional-quality video with clear demonstration of features and impact. **Good** (15–17): Well-made video with minor issues in clarity or engagement. **Satisfactory** (10–14): Basic video with limited engagement or unclear demonstration of features. **Needs Improvement** (0–9): Poor-quality video with little to no effort in showcasing the project.

University Policy on Late Submissions

If you submit coursework after the deadline, you will be penalized:

- **5%** of the total marks available for the assessment will be deducted from the assessment mark for **each working day** after the submission deadline, up to a maximum of 25%;
- Coursework received **more than five working days** after the submission deadline will receive a mark of **zero**.

University Policy on Academic Integrity

The University aims to foster a learning environment which produces students who embrace academic integrity, understand that they must produce their own work, are able to acknowledge explicitly any material that has been included from other sources or legitimate collaboration, and to present their own findings, conclusions or data based on appropriate and ethical practice.

The University will support you to understand the standards of academic integrity, while you are responsible for learning and upholding professional standards of research, writing, assessment, and ethics in your area of study. Violation of academic integrity comes in many forms, including but not limited to the following:

- improper citation or referencing;
- unauthorised collaboration with another person in the preparation and production of a submitted work;
- copying directly from other persons without their knowledge as your own work;
- submitting all or part of the same academic work for two or more modules without permission;
- consciously representing another's work or concept as your own without proper acknowledgment and citation of the sources;
- altering data obtained by legitimate means or making up a portion or whole set of data and reporting them in your own assignment;
- requesting another party to prepare all or part of an assignment (with or without payment) on your behalf.

Any violation of academic integrity is a serious offence and is therefore subject to an appropriate penalty. According to the individual case and the seriousness of the offence, penalties applied will vary and may include one or a combination of the following:

- a written warning;
- a mark penalty or a zero mark for the assessment;
- a zero mark for the module;
- a note on student's records;
- suspension of studies;
- termination of studies.

In addition to the respective penalty imposed, you may also be given feedback on how to avoid further offence in future work.